KnowSelf

Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Agentic Knowledgeable Self-Awareness for LLM-Based Agents

arxiv.org/abs/2504.03553

Current Agent Training Uses 'Flood Irrigation'

Indiscriminate Knowledge Injection Hurts Efficiency

Existing methods inject gold trajectories, external feedback, or domain knowledge at every step — regardless of whether the agent actually needs them.

Trajectory Imitation

e.g., ReAct

Blindly imitates gold action sequences without understanding when actions are needed.

- Fragile planning on unexpected inputs
- Pattern collapse under distribution shift

Trial-and-Error Refinement

e.g., Reflexion

Applies feedback at every step, even when the agent already knows the correct action.

- Unnecessary inference overhead
- Over-correction of correct steps

Knowledge-Augmented

e.g., KnowAgent, WKM

Injects domain knowledge at every step, even when the agent could act correctly on its own.

- High knowledge retrieval costs
- Knowledge interference on simple steps

Humans possess situational self-awareness — the metacognitive ability to assess whether they can handle a situation directly, need to reflect, or must seek external help. KnowSelf brings this to LLM agents.

Three Situational Modes

Fast Thinking · Slow Thinking · Knowledgeable Thinking

Decision Criterion: Predicted = Gold? → Fast | Reflection fixes? → Slow | Neither? → Knowledgeable

Fast Thinking

Criterion
Predicted action matches gold action on first attempt

Behavior
Agent directly produces the correct action with minimal deliberation

Token
No special token needed — direct action output

Slow Thinking

Criterion
Initial action is wrong, but self-reflection corrects it

Behavior
Agent pauses to reflect and self-correct before acting

Token
Generates Reflection token → triggers self-correction

Knowledgeable Thinking

Criterion
Agent cannot produce correct action even after reflection

Behavior
Agent consults external knowledge base for guidance

Token
Generates Knowledge token → triggers external KB retrieval

KnowSelf Framework

Data Construction → Two-Stage Training → Self-Aware Inference

1 Self-Awareness Data Construction

Agent self-explores the environment. At each step:

- Predicted = Gold → Fast token
- Reflection fixes it → Slow token
- Neither works → Knowledge token

Heuristic criterion classifies situations and inserts special tokens into trajectories.

2 Two-Stage Training

Stage 1: Supervised Fine-Tuning (SFT)
Teaches initial self-awareness patterns from token-labeled trajectories.

Stage 2: Reward-based Policy Optimization (RPO)
Refines the agent's ability to generate appropriate special tokens using reward signals.

Lightweight: only heuristic labeling on self-explored trajectories — no expensive human annotation.

3 Self-Aware Inference

During deployment, the agent generates special tokens that autonomously trigger:

- Direct action (fast thinking)
- Reflection on prior reasoning (slow thinking)
- Knowledge retrieval from external KB

No forced knowledge injection — the agent decides when external help is needed.

KnowSelf in Action

Correct reasoning where baselines fail — via reflection and knowledge tokens

Reflection

Task: "put a hot mug in cabinet"

OpenAI-O1 (fails):
Tries to take an egg from the microwave after opening it

KnowSelf (Llama-8B) (succeeds):
Uses Reflection token to realize it should heat the mug with the open microwave

Knowledge

Task: "put a saltshaker in drawer"

DeepSeek-R1 (fails):
Reflects that drawer is open, tries to close it first (fails)

KnowSelf (Llama-8B) (succeeds):
Applies Knowledge token: place object without removing unrelated items

Figure: Qualitative comparison — KnowSelf vs. OpenAI-O1 and DeepSeek-R1

ALFWorld Results

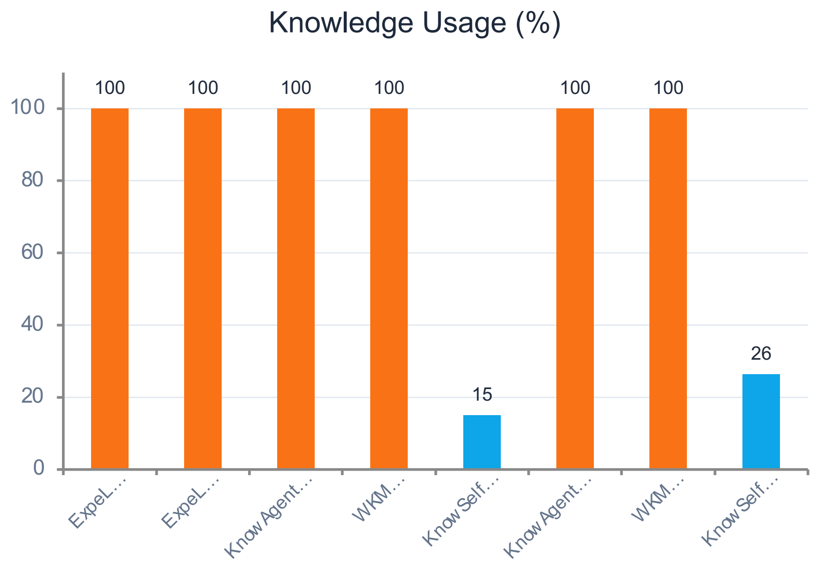
KnowSelf achieves 84.33% on ALFWorld with Llama-8B using only 15% knowledge

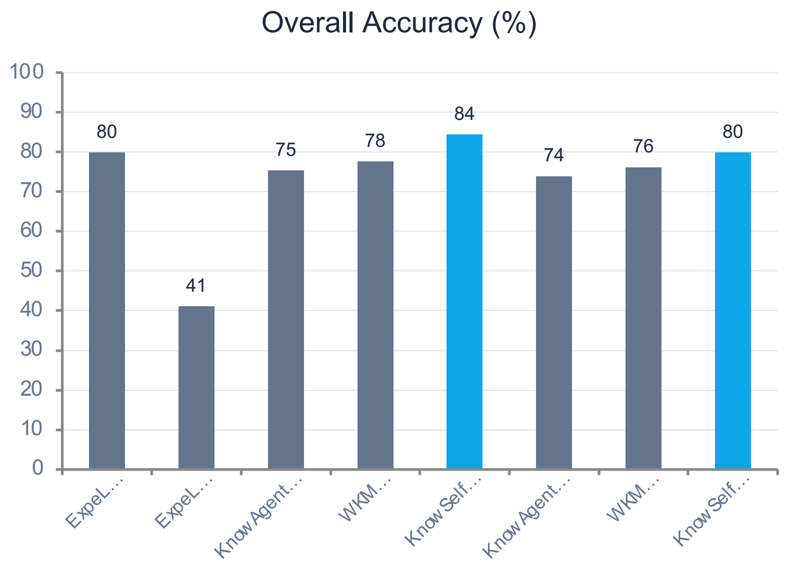
Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Two	All
GPT-4o	ReAct	0%	83.3	74.2	69.6	85.7	77.8	64.7	76.1
GPT-4o	Reflexion	0%	100	87.1	73.9	90.5	83.3	70.6	85.1
GPT-4o	ExpeL	100%	95.8	83.9	69.6	81.0	88.9	52.9	79.9
Llama-8B	ReAct	0%	33.3	3.2	0.0	57.1	66.7	23.5	27.6
Llama-8B	Reflexion	0%	63.0	22.7	5.9	64.3	86.4	50.0	51.5
Llama-8B	ExpeL	100%	83.3	32.3	30.4	23.8	55.6	17.7	41.0
Llama-8B	ETO	0%	91.7	70.6	82.6	61.9	88.9	64.7	78.4
Llama-8B	KnowAgent	100%	87.5	93.6	65.2	66.7	61.1	64.7	75.4
Llama-8B	WKM	100%	95.8	87.1	87.0	61.9	66.7	52.9	77.6
Llama-8B	**KnowSelf**	**15%**	**91.7**	**87.1**	**91.3**	**85.7**	**77.8**	**64.7**	**84.3**
Gemma-2B	ReAct	0%	0.0	9.7	0.0	4.8	44.4	0.0	9.0
Gemma-2B	ETO	0%	91.7	83.9	78.3	52.4	77.8	29.4	71.6
Gemma-2B	KnowAgent	100%	91.7	90.3	69.6	71.4	66.7	41.2	73.9
Gemma-2B	WKM	100%	91.7	87.1	78.3	71.4	61.1	52.9	76.1
Gemma-2B	**KnowSelf**	**26%**	**87.5**	**93.6**	**73.9**	**76.2**	**83.3**	**52.9**	**79.9**

Key: KnowSelf rows highlighted in blue. Methods using 100% knowledge (ExpeL, KnowAgent, WKM) often underperform KnowSelf's selective 15–26% usage.

Minimal Knowledge, Maximum Impact

KnowSelf's efficiency advantage over full-knowledge methods





Llama-8B: 15% knowledge → 84.33%

vs. KnowAgent 100% → 75.37% | WKM 100% → 77.61%

6–9 points higher with ~85% less knowledge

Gemma-2B: 26% knowledge → 79.85%

Matches GPT-4o ExpeL (79.85%) — a model 4× its size

Self-awareness generalizes across model scales

Key Takeaways

Self-Awareness Is the Missing Piece in Agent Planning

1. Indiscriminate knowledge injection is wasteful and suboptimal — flood-irrigation approaches hurt agent efficiency and performance.

2. Agents can learn situational self-awareness via special tokens with lightweight heuristic labeling on self-explored trajectories.

3. KnowSelf (Llama-8B) outperforms GPT-4o ReAct (84.33% vs 76.12%) and GPT-4o ExpeL (79.85%) on ALFWorld.

4. Only 15–26% knowledge usage needed versus 100% for competing methods — selective access outperforms blanket application.

5. The paradigm generalizes across model scales — even Gemma-2B reaches 79.85%, matching GPT-4o ExpeL.

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